Pollu-Sense: Abstract Report

AI-Based Air Quality Prediction and Monitoring System. Minor Project, B.Tech Information Technology, Poornima College of Engineering, Jaipur (RTU Kota), Session 2026-27.

Abstract

Pollu-Sense is an AI-based web platform that monitors air quality in real time, predicts pollution spikes before they occur, identifies the probable source of pollution and automatically alerts the parties responsible for corrective action. It addresses a clear gap in Indian cities: air quality is still measured through periodic reports from a limited number of fixed stations, so high pollution is detected but rarely attributed to a polluter, and action comes late.

The system collects data from Arduino-based IoT sensors, the OpenWeatherMap API and Central Pollution Control Board (CPCB) sources, and converts readings of PM2.5, PM10, NO_2, SO_2, CO and O_3 into the National Air Quality Index (AQI). A live dashboard and an interactive map show the current state, while TensorFlow and Scikit-learn models forecast future levels, detect abnormal spikes and classify the source as traffic, industry, construction or natural. When AQI crosses a configured threshold, an automated e-mail alert goes to the concerned industry or individual, and an unresolved case is escalated to the regulating authority.

The platform is built with React.js and Tailwind CSS on the front end, Django and Flask (with WebSockets) on the back end, PostgreSQL and Firebase for storage, authentication and notifications, and is hosted on Render. Crowdsourced reporting, a newsletter service, health-protection advice and an industry compliance scorecard extend it from simple monitoring to active pollution management. The project shows that AI, IoT and cloud services can be combined into a low-cost, scalable prototype for smart-city environmental governance, and it can be extended to more cities and pollution sources.

The main objectives of the work are to build a web-based monitoring platform with a live dashboard and map, to store sensor and public data for historical analysis, to apply machine-learning models for forecasting and spike detection, to attribute pollution to its likely source, to notify and escalate through an automated alert chain, and to encourage public participation through crowdsourced reporting and health guidance.

The development followed the complete software development life cycle: requirement analysis, system design, implementation, testing and deployment. The work was divided into six modules: data collection and sensor integration; back-end development and AI model training; front end and user interaction; real-time monitoring and alerts; industry compliance and historical tracking; and final deployment with Google Maps integration. The AI module runs as a separate service that reads from and writes to PostgreSQL, so new cities, sensors, data sources and models can be added without redesigning the core system. Testing covered individual modules, front-end to back-end integration, the end-to-end flow from data collection to alert delivery, and user acceptance of usability and responsiveness.

Key features of the system:

- Real-time AQI, temperature and humidity monitoring with a live dashboard and an interactive pollution map.
- AI-based prediction of future pollution levels and early detection of abnormal spikes.
- Source identification: traffic, industry, construction or natural causes.
- Automated e-mail alerts with an escalation path to regulatory authorities, including detection of repeat violations after an inspection.
- Industry compliance scorecard built from real-time and historical pollution records.
- Crowdsourced reporting, newsletter subscription and health-protection advice for citizens.
- IoT-ready design for emergency response, such as roadside mist sprays and air filters in high-pollution zones.

By linking measurement directly to action, Pollu-Sense gives citizens, industries and regulators a shared, transparent view of air quality. Its benefits include earlier preventive measures, lower exposure to harmful pollutants, reduced healthcare costs and productivity losses, and time-stamped evidence for regulatory intervention. The architecture is modular and low-cost, which makes it suitable for adoption by municipalities and smart-city programmes, starting with highly polluted cities.

Keywords

Air Quality Index (AQI), Air Pollution Prediction, Internet of Things (IoT), Machine Learning, Source Identification, Real-Time Monitoring, Automated Alerts, Crowdsourced Reporting, Smart City.

Problem and Motivation

Air pollution is one of the most serious public-health and environmental challenges in India, and current monitoring is periodic, sparse and reactive. Four gaps motivated the project:

- **No real-time monitoring:** pollution control boards depend on periodic reports from a limited number of fixed stations.
- **No attribution:** high levels are detected but not traced to a specific polluter.
- **Delayed action:** without instant alerts, industries and authorities respond late.
- **Public health burden:** long-term exposure raises the risk of asthma, bronchitis, COPD, lung cancer and cardiovascular disease, and affects children and the elderly most.

Indicator	Reported value	Source
Premature deaths linked to air pollution (global)	About 7 million per year	WHO
Indian cities exceeding safe air quality limits	Over 70%	CPCB report (2023)
Estimated annual economic loss to India	About $150 billion (about 3% of GDP)	Project study estimate
Share of urban air pollution from vehicles	About 30%	Project study estimate
Share of urban PM2.5 from construction and dust	20-30%	Project study estimate

Estimates differ between studies and methods; figures are as stated in the project report.

Proposed Approach

Pollu-Sense runs as a continuous cycle across five layers: sensing (IoT sensors, OpenWeatherMap, CPCB), data (cleaning and storage in PostgreSQL), intelligence (TensorFlow and Scikit-learn models), application (Django and Flask APIs with a React front end) and notification (alerts, e-mails and escalation reports).

1. **Monitor:** sensors and data feeds supply continuous readings for each region.
2. **Detect:** the AI model compares live readings with thresholds and learned patterns and flags an abnormal rise.
3. **Identify the source:** pattern analysis links the spike to traffic, industry, construction or a natural cause.
4. **Notify:** an automated e-mail with the AQI value, time and location goes to the concerned industry or individual.
5. **Escalate:** if no corrective action follows within the defined time, a report goes to the regulating authority.
6. **Predict:** collected data is fed back to the model to forecast the coming hours and days, so preventive steps can start early.

If an industry resumes polluting after clearing an inspection, the stored history lets the system detect the repeat violation and alert the authorities immediately. The platform serves three user groups: citizens (live AQI, health advice, reporting), industries (alerts and compliance score) and authorities (escalations, hotspots and history).

Technology Stack

Layer	Tools used	Role
Front end	React.js, Tailwind CSS, Chart.js, Google Maps API	Responsive dashboard, trend charts and interactive AQI map
Back end	Django, Flask with WebSockets	REST APIs, business logic and live push of readings and alerts
AI and ML	TensorFlow, Scikit-learn, pandas, NumPy	Prediction, spike detection and source classification
Data and storage	PostgreSQL	Sensor readings, predictions, alerts and historical records
Cloud services	Firebase Authentication and Cloud Messaging, Render	Sign-in, notifications and free-tier hosting
Data sources	IoT sensors (Arduino-based), OpenWeatherMap API, CPCB data	Local and public air quality and weather inputs

Outcomes, Conclusion and Future Scope

The working prototype delivers a home page with live AQI summary, an interactive map with location search and pollutant-wise readings, a dashboard with AQI, temperature and humidity cards and real-time trend charts, automated threshold alert e-mails, a newsletter with welcome and admin notifications, and a contact page. It was verified through unit, integration, system and user acceptance testing, including live dashboard updates, alert delivery and mobile responsiveness.

Pollu-Sense shows that AI, IoT and full stack web technologies can turn air quality monitoring from a passive, report-based activity into an alert-driven one that involves citizens, industries and regulators together. It reduces manual effort, improves transparency and gives authorities time-stamped evidence of pollution events and repeat violations.

Limitations: prediction and source attribution depend on the quality and coverage of sensor and historical data, and attribution is probabilistic, so field inspection should confirm it before any penalty. The prototype relies on third-party APIs and free-tier hosting, and is web-only.

Future scope: native mobile apps, satellite data, LSTM-based hybrid forecasting, blockchain-secured compliance records, a larger calibrated sensor network, deeper smart-city integration (traffic control, mist sprays and filters), gamified citizen participation, multi-language and accessibility support, regulator analytics, and expansion to more cities and pollution sources.

Objectives and Scope

Objectives:

- Design a web-based air quality platform that shows real-time pollution data on a dashboard and an interactive map.
- Collect data from IoT sensors, the OpenWeatherMap API and CPCB sources, and store it for historical analysis.
- Apply machine-learning models to predict pollution levels and detect abnormal spikes in advance.
- Identify whether pollution originates from traffic, industry, construction or natural causes.
- Send automated alerts to the responsible party and escalate unresolved cases to the authorities.
- Encourage public participation through crowdsourced reporting, a newsletter and health protection information.

Scope: real-time monitoring of AQI, temperature and humidity for multiple locations; prediction of near-future trends from historical and live data; alerts by e-mail, newsletter and a public contact channel; and a modular design that can start in highly polluted cities and later add cities, sensors and pollution sources.

Existing Practice vs Pollu-Sense

Aspect	Existing practice	Pollu-Sense approach
Monitoring	Periodic reports from a limited number of fixed	Continuous tracking using sensors, APIs and satellite-ready data feeds

Aspect	Existing practice	Pollu-Sense approach
	stations	
Prediction	Mostly reactive; levels reported after they occur	Machine-learning forecasts of spikes for preventive action
Source identification	High levels detected but not attributed to a polluter	AI analysis points to traffic, industry, construction or natural causes
Alerts and action	Delayed action and compliance issues	Automated e-mail alerts with escalation to authorities
Public participation	Citizens are only information consumers	Crowdsourced reporting, newsletter, reward ideas and health guidance
Compliance tracking	Manual inspections with little follow-up	Historical records and a compliance scorecard for each industry

Major Sources of Air Pollution

The literature survey identified six main sources, which the source-identification module is designed to separate:

- **Industrial emissions:** factories release carbon dioxide, sulphur dioxide and oxides of nitrogen.
- **Vehicular pollution:** transport contributes roughly 30% of urban air pollution.
- **Construction and dust:** about 20-30% of urban PM2.5.
- **Burning of fossil fuels:** coal-fired power plants are a major contributor.
- **Agricultural activities:** stubble burning sharply raises pollution, especially in North India.
- **Household pollution:** biomass fuels used for cooking add to indoor and outdoor pollution.

National Air Quality Index Used

Pollu-Sense converts pollutant concentrations into the CPCB National AQI so that readings are easy to understand and alert thresholds can be tied to categories.

AQI category	Range	Possible health impact
Good	0-50	Minimal impact
Satisfactory	51-100	Minor breathing discomfort to sensitive people
Moderately polluted	101-200	Discomfort to people with lung or heart disease, children and older adults
Poor	201-300	Breathing discomfort to most people on prolonged exposure
Very poor	301-400	Respiratory illness on prolonged exposure

AQI category	Range	Possible health impact
Severe	401-500	Affects healthy people and seriously affects those with existing diseases

Related Work and Regulatory Context

Reviews of machine-learning techniques for air pollution prediction (Sulaimon et al.) and for PM2.5 forecasting (Palanichamy et al.) show that data-driven models can forecast pollutant concentrations from historical and meteorological data, which supports the prediction module. A review of IoT sensor nodes for air pollution monitoring (Guerron et al.) motivates the sensor-based data collection layer. Further studies on AI-assisted monitoring for smart cities, low-cost IoT monitoring, crowdsourced pollution data, and AI-assisted atmospheric modelling for faster forecasts informed the integrated design.

The Air (Prevention and Control of Pollution) Act, 1981 gives the Central Pollution Control Board a leading role in controlling air pollution, and the Environment (Protection) Act, 1986 empowers the Government of India to control pollution from all sources and restrict industrial facilities on environmental grounds. Pollu-Sense supports this framework by supplying timely, time-stamped evidence of pollution events and repeat violations to the regulators.

System Architecture and Users

Layer	Responsibility
Sensing	IoT sensors and external APIs (OpenWeatherMap, CPCB) supply raw air quality and weather data
Data	Data is cleaned, validated and stored in PostgreSQL with historical records
Intelligence	TensorFlow and Scikit-learn models predict levels, detect spikes and classify the probable source
Application	Django and Flask services expose APIs and WebSockets to the React front end
Notification	Alerts, e-mails, newsletters and escalation reports are generated and delivered

User	Main use cases
Citizen	View live AQI and map, read health advice, subscribe, report pollution
Industry	Receive alerts, take corrective action, view own compliance score
Authority	Receive escalations, view history and hotspots, decide regulatory action
AI module	Detect spikes, identify sources, predict levels, trigger alerts

Implementation Modules

1. **Data collection and sensor integration:** data from the OpenWeatherMap API, CPCB data sets and IoT sensors.
2. **Back end and AI model training:** Django APIs; TensorFlow and Scikit-learn models for prediction and source classification.
3. **Front end and user interaction:** React.js site with Firebase authentication, dashboard, interactive map, newsletter and crowdsourced reporting.
4. **Real-time monitoring and alerts:** Flask API with WebSockets, Firebase Cloud Messaging and e-mail notifications.
5. **Industry compliance and historical tracking:** patterns stored in PostgreSQL, visualised with Chart.js, with a compliance rating per industry.
6. **Deployment and smart-city integration:** hosting on Render (free tier) with Google Maps API for visualisation.

Testing

Test scenario	Expected behaviour
Live data reaches the dashboard	AQI, temperature and humidity cards update without a page reload
AQI crosses the configured threshold	An alert e-mail with the AQI value is sent to the subscriber
A user subscribes to the newsletter	A welcome e-mail is received and the subscription is recorded
A location is searched on the map	The map moves to the location and shows its AQI details
The site is opened on a phone	The layout adapts to the screen size and stays readable

Recommendations

- Deploy a wider, regularly calibrated sensor network to improve coverage and reliability.
- Use deep-learning and hybrid models trained on more historical data.
- Keep alerts, inspection results and compliance scores in a tamper-proof audit trail.
- Add a mobile app and a multilingual interface (Hindi and regional languages) to reach more citizens.
- Strengthen the back end for large-scale, multi-city deployment.